Label symmetry is necessary for the Magniez–Zur collision tradeoff

Abstract

For a uniformly random function on N labels, we construct a quantum collision algorithm with average success probability at least $2/3$, total retained space $S = O(\log N)$, and query complexity $T = O(\sqrt{N \log\log N / \log N})$. Thus $T^2 S = O(N \log\log N)$, showing that the Magniez–Zur lower bound for label-symmetric algorithms does not extend to unrestricted value-oracle algorithms. The construction combines numerical fingerprints with quantum search.

For each integer N, choose a function $f : [N] \to [N]$ uniformly from all N^N functions. Its coherent value oracle acts by

$$O_f|x, y\rangle = |x, y \oplus \mathrm{enc}(f(x))\rangle.$$

The algorithm is fixed before f is chosen. Its success probability is averaged over that one function and its measurement outcomes. The resource T counts oracle queries; S counts all retained classical bits and qubits, including workspace and output registers.

Theorem. The bound $T^2 S = \Omega(N \log N)$ in the unrestricted random-function query/space model is false. For every sufficiently large N, there is an algorithm with unconditional average success at least $2/3$, a fixed number of coherent value-oracle queries, and

$$S = O(\log N), \qquad T = O\left(\sqrt{\frac{N \log\log N}{\log N}}\right).$$

Thus

$$T^2 S = O(N \log\log N) = o(N \log N).$$

All logarithms in parameter definitions are base two. Identify $[N]$ with $\{0, \ldots, N-1\}$ through fixed input/output encodings. Such a fixed relabelling changes no collision relation and adds only input-independent interface gates and $O(\log N)$ scratch if needed.

Magniez and Zur [MZ26, Theorem 4.19] prove $T^2 S = \Omega(N \log N)$ for label-symmetric algorithms and leave the unrestricted case open in their Section 1.4. The construction below uses numerical fingerprints of output labels and shows that the same bound does not extend to arbitrary value-oracle algorithms. The query and space resources have the same interpretation: intermediate operations are input-independent, and all retained data are charged. If a collision output is required to include its common value, retain that value from the final verification query; this adds at most n bits to the space bound and leaves the asymptotic separation unchanged.

The construction follows the sample-and-search approach of Brassard, Høyer and Tapp [BHT97], using Grover search [G96]. It stores numerical fingerprints in place of full sample values and checks matches using the oracle.

1 A parameterized algorithm

First let r be any integer with $r \geq 1024$ and $N \geq 64r^2$. Write

$$n = \lceil \log_2 N \rceil, \quad b = \lceil \log_2(64r^2) \rceil, \quad B = 2^b,$$
$$A = \{0, \ldots, r-1\}, \quad D = \{r, \ldots, N-1\}, \quad L = N - r.$$

Because $N \geq 64r^2$, $b \leq n$. Define the fixed numerical fingerprint

$$h(y) = y \bmod B.$$

For each fixed sample input $i \in A$, store only

$$a_i = h(f(i))$$

in a b-bit table slot. The corresponding original input index is its fixed slot; no full sample values or n-bit index array are retained. Each entry uses two queries: load $f(i)$ into an n-bit value register, copy its low b bits to the slot, and erase the full value by an inverse query. This uses $2r$ queries.

For an XOR value oracle the inverse is the oracle itself. For addition modulo N, the inverse uses one forward query conjugated by the fixed response negation $J|y\rangle = |-y \bmod N\rangle$. Fixed output encodings can likewise be converted before copying the tag and converted back before unloading the value.

More explicitly, an XOR query can be simulated using the addition-modulo-N oracle of [MZ26]. On a valid address x, load $f(x)$ into a zeroed $\lceil \log_2 N \rceil$-qubit temporary register, XOR its binary encoding into the response register, and erase the temporary register with an inverse addition query. The preceding negation identity implements that inverse with one forward query. Thus each XOR query costs two addition queries and $O(\log N)$ extra workspace; the asymptotic bounds are unchanged.

Compute a retained bit F indicating that all a_i are distinct. This requires no queries and can be done reversibly with small scratch: count equal-tag pairs in a counter of width $\lceil \log_2(\binom{r}{2} + 1) \rceil$, using one equality flag at a time; toggle F iff the count is zero; reverse the scan to erase the counter. Equality testing and conditional increments use $O(b + \log r)$ clean scratch. The table is preserved. F and all counter/controller bits are charged.

For a full value y, a reversible sequential table scan computes

$$J(a, y) = \bigoplus_{\{i : a_i = h(y)\}} i, \qquad C(a, y) = \#\{i : a_i = h(y)\} \bmod 2.$$

The empty XOR is zero. On $F = 1$, these are exactly the unique matching original sample index and its existence bit.

The scan consists of reversible gates on the stored table. For each fixed i, compute the equality flag $[a_i = h(y)]$ with $O(b)$ clean scratch, conditionally XOR the fixed binary representation of i into the output index and toggle the output bit, then erase the equality flag. Reverse the scan to erase the index and bit when desired. A live loop index costs $O(\log r)$ charged bits; an unrolled circuit has the same input-independent schedule. Every function-dependent table bit remains a data qubit.

Even when $F = 0$, the XOR index is valid:

$$0 \le J < 2^{\lceil \log_2 r \rceil} \le 2r < N.$$

Thus no bad-table or absent-tag branch requests an invalid oracle address.

Define, on D, the phase predicate

$$P_f(x) = [F = 1]\,[C(a, f(x)) = 1]\,[f(J(a, f(x))) = f(x)].$$

When $F = 1$, this is exactly membership of $f(x)$ in the r-element set $\{f(i) : i \in A\}$. When $F = 0$, it is identically zero.

A clean phase marker costs four queries:

1. Load $f(x)$ into Y.
2. Compute J, C by the table scan.

3. Load $f(J)$ into Z.
4. Apply the sign conditioned on F, C and the full n-bit equality $Y = Z$.
5. Erase Z with an inverse query, reverse the table scan, and erase Y with an inverse query.

All scratch is returned to zero. If there is no matching tag, $J = 0, C = 0$; querying the valid address zero and uncomputing it is harmless. The full-value comparison prevents a fingerprint false positive from being accepted.

The coherent value oracle acts through a single query interface, implemented by fixed swaps or by copying X or J into a charged temporary address Q, making the query, and erasing Q. If the physical response port is Y, place Z there by swapping Y, Z before and after each query intended for Z; this preserves the other full value and adds no register or query. These registers are included in the space bound below.

Prepare

$$|u_D\rangle = L^{-1/2} \sum_{x \in D} |x\rangle$$

and its reflection. These are input-independent unitaries of width $O(n)$, including scratch. An explicit preparation uses binary prefix rotations: for each prefix p, its weight $w(p)$ is the number of elements of the fixed interval D having that prefix, and the next-bit amplitudes are $\sqrt{w(p0)/w(p)}$ and $\sqrt{w(p1)/w(p)}$. Zero-weight prefixes use arbitrary fixed rotations. The rotations can be arranged on the address bits with $O(n)$ scratch; their gate count does not enter the query complexity. The reflection is the preparation, a zero-state reflection, and inverse preparation.

Run exactly

$$m = \left\lfloor \frac{3}{4}\sqrt{L/r} \right\rfloor = \left\lfloor \sqrt{\left\lfloor \frac{9L}{16r} \right\rfloor} \right\rfloor$$

Grover iterations. This integer is determined by N, r, never by the unknown marked count. Computing or retaining it and all control counters costs $O(n)$ bits.

After measuring x, make two further value queries to compare its full value with the selected sample value. On a marked result the measured pair (J, x) consists of distinct valid indices with equal full oracle values. The success event consists of distinct measured addresses with equal oracle values; all other outcomes count as failures.

The query count is identical on every function:

$$T = 2r + 4m + 2 \le 2r + 3\sqrt{L/r} + 2.$$

2 Unconditional success probability

A fingerprint bucket has at most $\lceil N/B \rceil$ values, even when N is not a multiple of B. For independent uniform $U, V \in [N]$, if $q_t = \Pr(h(U) = t)$,

$$\Pr(h(U) = h(V)) = \sum_t q_t^2 \le \max_t q_t \le 1/B + 1/N.$$

Let $E = \{F = 1\}$. Independence of the sample coordinates and the union bound give

$$\Pr(E) \ge 1 - \binom{r}{2}(1/B + 1/N) \ge 63/64.$$

Condition on any complete sample vector satisfying E. Its full values are distinct. The same fixed oracle's outside coordinates $f(x)$, $x \in D$, are still independent uniform values. If K counts marked search inputs, then

$$K \sim \mathrm{Bin}(L, r/N), \qquad \mu = r(1 - r/N), \qquad \mathrm{Var}\, K \le r.$$

This law also holds conditional only on E, since its parameter is the same for every such complete sample.

Let $G = \{|K - \mu| \le r/8\}$. Chebyshev yields

$$\Pr(G \mid E) \ge 1 - 64/r \ge 15/16.$$

On G, using $r^2/N \le 1/64$,

$$3r/4 \le K \le 5r/4.$$

For a fixed oracle, put

$$a = \sqrt{r/L} \le 1/10, \qquad \theta = \arcsin\sqrt{K/L}, \qquad q = \theta/a.$$

The lower count bound gives $q \ge \sqrt{3/4} > 5/6$. The elementary integral inequality $\arcsin u \le u/\sqrt{1-u^2}$ gives

$$q \le \sqrt{\frac{5/4}{1 - (5/4)a^2}} \le \sqrt{100/79} < 7/6.$$

Thus $q \in [5/6, 7/6]$.

In the marked/unmarked two-dimensional subspace, the initial state has coordinates $(\sin\theta, \cos\theta)$, and one Grover iteration sends them to $(\sin 3\theta, \cos 3\theta)$. After m iterations, the probability of measuring a marked input is therefore $\sin^2((2m+1)\theta)$.

For the prescribed rational iteration count there is an e with $|e| \le 1$ such that

$$a(2m+1) = 3/2 + ea.$$

Since $3q/2 \in [5/4, 7/4]$ and $3 < \pi < 22/7$,

$$|(2m+1)\theta - \pi/2| \le 9/28 + 7/60 = 46/105 < \pi/6.$$

Grover success on $E \cap G$ is at least $3/4$. All such marked outputs pass the exact final verification. The unconditional average success over f and measurement is consequently at least

$$\frac{63}{64}\frac{15}{16}\frac{3}{4} = \frac{2835}{4096} > \frac{2}{3}.$$

3 Space complexity

The retained fingerprint table uses rb qubits/bits. Main and selected addresses X, J, the physical address Q if used, and the two full value registers Y, Z use at most $5n$ qubits. The pair-count check and sequential scan use $O(b + \log r)$ scratch/counter bits. Full equality, exact interval-state preparation, classical arithmetic, loop and iteration controls, fixed parameter records and output registers each add $O(n)$. Scratch can be reused, but even summing these allocations gives

$$S \le rb + Cn$$

for an absolute constant C, since $b \le n$ and $\log r \le n$.

The space bound counts classical bits and qubits equally, including the table and the two retained output indices. As usual in the query model, the supplied oracle's internal storage is excluded.

Thus the parameterized costs are

$$T = O(r + \sqrt{N/r}), \qquad S = O(\log N + r \log r).$$

4 Choosing the parameters

Let

$$\ell = \lceil \log_2 n \rceil, \qquad r = \lfloor n/\ell \rfloor$$

for sufficiently large N. Then $r \to \infty$, $r^3 = o(N)$, and $N \ge 64r^2$; hence all the preceding hypotheses eventually hold. Also $r \asymp n/\log n$, and

$$rb \le (n/\ell)(7 + 2\log_2 n) = O(n).$$

All charged workspace is therefore $S = O(n)$. One explicit sufficient threshold is $N \ge 2^{65536}$: then $n \ge 65536$, $\ell \ge 16$, $r \ge 2048$, and $N > 2^{n-1} \ge 64n^2 \ge 64r^2$. Also $rb \le 2n + 7n/\ell < 3n$. Thus the construction applies to every sufficiently large integer N, without a power-of-two restriction. Since preprocessing is negligible,

$$T = O\left(\sqrt{N\ell/n}\right).$$

It follows that

$$T^2 S = O(N\ell) = O(N \log\log N), \qquad \frac{T^2 S}{N \log N} = O(\ell/n) \longrightarrow 0.$$

Consequently, the bound $T^2 S = \Omega(N \log N)$ cannot hold for all algorithms in the stated oracle model.

5 References

- **[MZ26]** F. Magniez and S. Zur, *Tight Time-Space Lower Bounds for Collision Finding and Element Distinctness under Label Symmetry* (2026), arXiv:2609.10808, Definition 4.4, Theorem 4.19 and Section 1.4.
- **[BHT97]** G. Brassard, P. Høyer and A. Tapp, *Quantum Algorithm for the Collision Problem* (1997), arXiv:quant-ph/9705002.
- **[G96]** L. K. Grover, *A fast quantum mechanical algorithm for database search* (1996), arXiv:quant-ph/9605043.